Mohimenul Kabir (Mahi)

PhD Candidate, National University of Singapore
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Research interests

Automated Reasoning, Formal Methods, Logic Programming, Answer Set Programming, Artificial Intelligence, Metaheuristic

Education

2020 – **Present** **National University of Singapore** (NUS) – Singapore
PhD in Computer Science
Advisor: Kuldeep S. Meel.

2013 – 2017 **Bangladesh University of Engineering and Technology** (BUET) – Bangladesh
BSc in Computer Science
Mentors: M Sohel Rahman.

Honors and scholarships

2023 NUS SoC Research Incentive Award

2020 NUS Research Scholarship

2014 & 2015 University Merit Scholarship (Bangladesh University of Engineering and Technology)

Publications

2025 **An ASP-Based Framework for MUSes**
Mohimenul Kabir and Kuldeep S Meel
To appear ICLP 2025.

2025 **Scalable Counting of Minimal Trap Spaces and Fixed Points in Boolean Networks**
Mohimenul Kabir, Van-Giang Trinh, Samuel Pastva, and Kuldeep S Meel
To appear CP 2025.

2024 **Image Contrast Enhancement using Fuzzy Logic with Metaheuristics**
Best Student Paper award of NSysS 2024
Mohimenul Kabir, Jaiaid Mobin, Ahmad Hassanat, and M. Sohel Rahman.
NSysS 2024.

2024 **On Lower Bounding Minimal Model Count**
Best Paper award of ICLP 2024
Mohimenul Kabir and Kuldeep S Meel
Theory and Practice of Logic Programming (Special issue of ICLP 2024).

2024 **Exact Answer Set Counting with Compact Encoding**
Mohimenul Kabir, Supratik Chakraborty and Kuldeep S Meel
Proceedings of the AAAI 2024 (**Invited in MCW Workshop 2024**).

2023 **A Fast and Accurate ASP Counting Based Network Reliability Estimator**
Mohimenul Kabir and Kuldeep S Meel
LPAR 2023. (**Invited in MCW Workshop 2023**)

2023 **Multi-objective optimization and heuristic based solutions for evacuation modeling**
Mohimenul Kabir, Jaiaid Mobin, Muhammad Ali Nayeem, Muhammad Ahsanul Habib, and M. Sohel Rahman
Transportation Research Interdisciplinary Perspectives.

2022 **ApproxASP – A Scalable Approximate Answer Set Counter**
Mohimenul Kabir, Flavio O Everardo, Ankit K Shukla, Markus Hecher, Johannes Klaus Fichte, and Kuldeep S Meel
Proceedings of the AAAI. (**Invited in ICLP 2022 and Indian SAT-SMT school 2022**)

2020 **Evaluation of classification and forecasting methods on time series gene expression data**
Nafis Irtiza Tripto, **Mohimenul Kabir**, Md Shamsuzzoha Bayzid, Atif Rahman
PLOS ONE.

2020 **ADACT: a tool for analysing (dis) similarity among nucleotide and protein sequences using minimal and relative absent words**
Mujtahid Akon, Muntashir Akon, **Mohimenul Kabir**, M Saifur Rahman, M Sohel Rahman
Bioinformatics.

2018 **A simple, fast, filter-based algorithm for circular sequence comparison**
Md Azim, Aashikur Rahman, **Mohimenul Kabir**, M Sohel Rahman
In International Workshop on Algorithms and Computation (WALCOM).

Research Visits

Oct 2024 **Texas Tech University**

Jan – June 2024 **University of Toronto**

Teaching experience

Fall 2023 **Teaching assistant, IT5005: Introduction to Artificial Intelligence (NUS)**

Spring 2021,23 **Teaching assistant, CS4244: Knowledge Representation and Reasoning (NUS)**

Fall 2021 **Teaching assistant, CS3243: Introduction to Artificial Intelligence (NUS)**

Spring 2022 **Teaching assistant, CS4246/CS5446: AI Planning and Decision Making (NUS)**

Fall 2022 **Teaching assistant, CS2040: Data Structures and Algorithms (NUS)**

Talks

December 2024	Image Contrast Enhancement using Fuzzy Logic with Metaheuristics *NSysS 2024*
October 2024	On Lower Bounding Minimal Model Count *ICLP 2024*
October 2024	Answer Set Counting and its Applications *ICLP LPNMR Doctoral Consortium 2024*
August 2024	Exact ASP Counting with Compact Encodings *Model Counting and Sampling workshop 2024*
February 2024	Exact ASP Counting with Compact Encodings *AAAI 2024*
June 2023	A Fast and Accurate ASP Counting Based Network Reliability Estimator *LPAR 2023*
December 2022	ApproxASP – a scalable approximate answer set counter *SAT SMT Winter School 2022*
February 2022	ApproxASP – a scalable approximate answer set counter *AAAI 2022*

Mentorship and service

May – July 2022 **National University of Singapore (Academic Advisor)**
My responsibilities: I mentored two undergraduate students on a research project. The title of the project: "Approximate Answer Set Sampling"

Technical skills

Programming languages: C, C++, Java, Python, R

References

Kuldeep S Meel (meel@cs.toronto.edu, University of Toronto)

Supratik Chakraborty (supratik@cse.iitb.ac.in, IIT Bombay)

Johannes K. Fichte (johannes.fichte@liu.se, Linkoping University)